05/05/26

1.
```
C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 16>javac --release 8 *.java
warning: [options] source value 8 is obsolete and will be removed in a future release
warning: [options] target value 8 is obsolete and will be removed in a future release
warning: [options] To suppress warnings about obsolete options, use -Xlint:-options.
3 warnings

C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 16>java Part1Main
=== TryThreeTimes ===
Attempt 1: 71
You succeeded

=== TryForEver ===
Attempt 1: 0.5561330981284616
Attempt 2: 0.8760042371460843
Attempt 3: 0.11144376666872069
You succeeded
```

2c. NestedLoopsComputation.java:

```
C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 16>java NestedLoopsComputation
n = 1000
Nested loop time: 6 ms

n = 5000
Nested loop time: 33 ms

n = 10000
Nested loop time: 70 ms
```

ThreadedPoolConputation.java:

```
C:\Users\vgmat\Downloads\Principles of Programming Concepts\Week 16>java ThreadPoolComputation
Cores available: 4

n = 1000
Thread pool time: 82 ms

n = 5000
Thread pool time: 10 ms

n = 10000
Thread pool time: 31 ms
```

Hand calculation for n=3:

$n=3$

$a=[1,2,3]$

$b^1=[4,5,6]$

$b^2=[7,8,9]$

$b^3=[1,2,3]$

$$a_i \cdot b_i^1 = (1)(4)+(2)(5)+(3)(6)$$
$$= 4+10+18$$
$$= 32$$

$$a_i \cdot b_i^2 = (1)(7)+(2)(8)+(3)(9)$$
$$= 7+16+27$$
$$= 50$$

$$a_i \cdot b_i^3 = (1)(1)+(2)(2)+(3)(3)$$
$$= 1+4+9$$
$$= 14$$